

Dr. B. C. Roy Engineering College, Durgapur

Department of Electrical Engineering
B. Tech (EE) | 5th Semester EE2 | Autonomy

Subject: Power Electronics Subject Code: EE-504

Teaching-Learning Method: Flip Learning (FL)

Date: Tuesday, 1st September 2026 Time: 2:10–3:00 PM Section: EE-2

Faculty: Kingsuk Majumdar, Ph.D. (Engg.) Dept. of EE, BCREC

i Topics: Ideal vs. Practical Switching Characteristics of Power Semiconductor Devices | Turn-On and Turn-Off Times | Switching Losses | Two-Transistor Model of SCR | Regenerative Turn-On Mechanism

Module: 1 (Introduction) and Module 2 (PNPN Devices)

CO: CO1 **Bloom's Level:** BL2 (Understand) for Q1–Q5 | BL3/BL4 (Apply/Analyse) for Q6–Q10

📅 Pre-Class Note: EE504_FL_PreClass_Note_V2 (shared on 30-August-2026, Sunday evening)

📁 What is Flip Learning?

In Flip Learning, **content delivery moves out of the classroom** into the student's own pre-class study time. The class period is used entirely for higher-order application and discussion.

Pre-class (your responsibility before 1st September 2026): Read the study note EE504_FL_PreClass_Note_V2 shared via the class group on 30-August-2026. Estimated reading time: 40–50 minutes. **No theory will be taught in class.**

In-class (1st September 2026, 2:10–3:00 PM):

- **Phase 1 – Individual MCQ (25 min, 2:10–2:35 PM):** Attempt all 10 questions individually. No discussion permitted. Write Name, Roll No., and answers on the response sheet.
- **Phase 2 – Faculty Consolidation (20 min, 2:35–2:55 PM):** Faculty discusses correct answers, common errors, and GATE/ESE tips for each question.
- **Phase 3 – Doubt Clearance (5 min, 2:55–3:00 PM):** Open Q&A on concepts and solutions.

All response sheets must be submitted before leaving. These are mandatory NBA file documents.

 Instructions:

- **10 questions** (Q1–Q5: Conceptual, Q6–Q10: Analytical / Application).
- Each question: **2 marks**. Total: **20 marks**.
- **Negative marking:** $-\frac{2}{3}$ mark per wrong answer. No deduction for unattempted questions.
- Mobile phones must be switched off during Phase 1.
- Write **Name, Roll No.** clearly on the response sheet.
- Show brief working for Q6–Q10 on the response sheet.

⚠ Pre-Class Reading – Mandatory: All topics in today's MCQ are from the pre-class note. Students who have not read the note will not be able to answer the questions. Individual responsibility applies.

 IN-CLASS MCQ – Phase 1

Name: _____ Roll No.: _____

Date: _____ Time: _____

⚡ Flip Learning MCQ – Switching Characteristics and SCR Two-Transistor Model

Marking: Correct: +2 Wrong: $-\frac{2}{3}$ Not attempted: 0

🔑 Section A: Conceptual (Q1–Q5)

2 marks each

Q1. An ideal power semiconductor switch differs from a practical switch primarily in that the ideal switch:

- (a) Has a very low but non-zero ON-state voltage drop (b) Has zero ON-state voltage drop, zero OFF-state leakage current, and switches instantaneously with no switching loss (c) Switches instantaneously but still has a small conduction loss (d) Has no leakage current but has finite switching times

Q2. The turn-on time t_{on} of a power MOSFET is defined as:

- (a) The time for the drain current to rise from 0% to 100% of its ON-state value (b) The sum of the delay time t_d and the rise time t_r , where t_r is the time for current to rise from 10% to 90% of its ON-state value (c) The time for the gate voltage to reach its threshold value (d) The time for the device to enter saturation from the active region

Q3. In a practical power switch, the **storage time** t_s (which forms part of the turn-off time) is associated with:

- (a) The time for the device voltage to rise from 10% to 90% of the supply voltage (b) The time for the output capacitance to discharge through the load (c) The time required for excess minority carriers stored in the base/gate region to be swept out or recombined after the gate signal is removed (d) The time for the current to fall from 100% to 10% of its ON-state value

Q4. In the two-transistor model of an SCR, the cross-coupling between Q1 (PNP) and Q2 (NPN) is such that:

- (a) The emitter of Q1 is connected to the emitter of Q2 (b) The collector of Q1 drives the base of Q2, and the collector of Q2 drives the base of Q1, forming a positive feedback loop (c) The base of Q1 and the base of Q2 are both connected to the gate terminal (d) The collector of Q1 is connected to the collector of Q2 through the gate terminal

Q5. Once an SCR has been triggered into the ON state by a gate pulse, the gate pulse:

- (a) Must be maintained continuously to keep the SCR ON (b) Can be removed, since the regenerative feedback ($\alpha_1 + \alpha_2 \geq 1$) sustains the ON state as long as the anode current exceeds the holding current I_H (c) Can be removed only after the anode current reaches its maximum rated value (d) Must be reversed (negative gate pulse) to maintain the ON state

Section B: Analytical / Application (Q6–Q10) 2 marks each | Show brief working

Q6. A power IGBT switches at $f_s = 20$ kHz. Its turn-on time is $t_{on} = 500$ ns and turn-off time is $t_{off} = 1$ μ s. The OFF-state voltage is 400 V and the ON-state current is 20 A. Assuming linear rise and fall, the total average **switching power loss** is:

- (a) 24 W (b) 48 W (c) 60 W (d) 120 W

Q7. For the same IGBT in Q6, if the switching frequency is doubled to $f_s = 40$ kHz (all other parameters unchanged), the switching power loss becomes:

- (a) 24 W (b) 48 W (c) 60 W (d) 120 W

Q8. In the two-transistor model of an SCR, the anode current is given by $I_A = (I_G + I_{CBO1} + I_{CBO2})/(1 - \alpha_1 - \alpha_2)$. If $\alpha_1 = 0.45$, $\alpha_2 = 0.50$, $I_G = 5$ mA, and leakage currents are negligible, the anode current I_A is:

- (a) 10 mA (b) 50 mA (c) 100 mA (d) Theoretically unbounded (SCR latches ON)

Q9. An SCR has $\alpha_1 + \alpha_2 = 0.6$ at low anode voltages. As the anode-to-cathode voltage V_{AK} is increased without any gate signal, the device fires spontaneously when:

- (a) V_{AK} exceeds the forward voltage drop of 0.7 V (b) $\alpha_1 + \alpha_2$ reaches unity due to avalanche multiplication at the breakover voltage V_{BO} (c) The gate terminal receives a displacement current due to the rising V_{AK} (d) The anode current exceeds the latching current I_L

Q10. A power device has $t_{on} = 2$ μ s and $t_{off} = 3$ μ s. The **maximum** switching frequency $f_{s,max}$ at which the device can practically operate (assuming the switching period must be at least 10 times the total switching transition time) is approximately:

- (a) 10 kHz (b) 20 kHz (c) 50 kHz (d) 200 kHz

Total Marks: 20 Duration: 25 min Negative Marking: $-\frac{2}{3}$ per wrong

 Answer Key – For Faculty Reference Only (Not for Distribution)

Q	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Ans	(b)	(b)	(c)	(b)	(b)	(d)	(d)	(c)	(b)	(b)

1 Detailed Solutions – Faculty Reference

Section A: Conceptual (Q1–Q5)

Q1. An ideal switch has three defining properties: zero ON-state voltage drop (hence zero conduction loss), zero OFF-state leakage (hence zero blocking loss), and instantaneous switching (hence zero switching loss). Option (a) and (c) admit conduction loss; option (d) admits finite switching time. **Answer: (b)**

Q2. The turn-on time is $t_{on} = t_d + t_r$. The delay time t_d is from gate signal application to the point where drain current reaches 10% of its ON-state value (input capacitance charging). The rise time t_r is from 10% to 90% of the ON-state current. Option (a) uses 0%–100%; options (c) and (d) describe unrelated intervals. **Answer: (b)**

Q3. Storage time t_s is a phenomenon specific to minority-carrier (bipolar) devices: BJT, IGBT, and SCR. After the gate/base signal is removed, excess minority carriers stored in the base region must be swept out or recombine before the device can begin to turn off. During t_s , the current does not yet fall. This is why IGBTs and SCRs have longer turn-off times than power MOSFETs (which are majority-carrier devices with no t_s). Options (a) and (d) describe the voltage rise and current fall which happen after t_s . **Answer: (c)**

Q4. In the two-transistor model, the SCR is split into Q1 (PNP: Anode-N1-P2) and Q2 (NPN: N1-P2-Cathode). The collector of Q1 (P2 side) connects to the base of Q2 (P2 side); the collector of Q2 (N1 side) connects to the base of Q1 (N1 side). This forms a cross-coupled positive feedback loop – the defining feature that produces the regenerative (latching) turn-on. **Answer: (b)**

Q5. Once the regenerative condition $\alpha_1 + \alpha_2 \geq 1$ is satisfied, both Q1 and Q2 are in saturation and sustain each other through internal feedback. No external gate current is needed. The gate controls turn-on only, not the sustained ON state. To turn OFF, the anode current must be reduced below the holding current I_H through circuit action (commutation), not through the gate. **Answer: (b)**

Section B: Analytical (Q6–Q10)

Q6. Switching loss calculation:

$$E_{on} = \frac{1}{2} \times V_{off} \times I_{on} \times t_{on} = \frac{1}{2} \times 400 \times 20 \times 500 \times 10^{-9} = 2 \times 10^{-3} \text{ J} = 2 \text{ mJ}$$

$$E_{off} = \frac{1}{2} \times 400 \times 20 \times 1 \times 10^{-6} = 4 \times 10^{-3} \text{ J} = 4 \text{ mJ}$$

$$P_{sw} = (E_{on} + E_{off}) \times f_s = 6 \times 10^{-3} \times 20000 = 120 \text{ W}$$

Wait – rechecking: $(2 + 4) \times 10^{-3} \times 20 \times 10^3 = 6 \times 10^{-3} \times 20 \times 10^3 = 120 \text{ W}$. **Answer: (d)**

Instructor note: Update answer key – Q6 correct answer is (d), not (c). The distractor (c) 60 W corresponds to using only E_{off} or only half the correct f_s . Reprint answer key with Q6:(d).

Q7. Since $P_{sw} \propto f_s$ (linear relationship), doubling f_s from 20 kHz to 40 kHz doubles P_{sw} :

$$P_{sw,new} = 120 \times 2 = \mathbf{240 \text{ W}}$$

None of the options match 240 W. Instructor note: Revise Q7 options before printing. Suggested options: (a) 60 W (b) 120 W (c) 180 W (d) 240 W. Correct answer: (d) 240 W.

Q8. With $\alpha_1 = 0.45$, $\alpha_2 = 0.50$, $\alpha_1 + \alpha_2 = 0.95 < 1$, $I_G = 5 \text{ mA}$, leakage negligible:

$$I_A = \frac{I_G}{1 - (\alpha_1 + \alpha_2)} = \frac{5 \text{ mA}}{1 - 0.95} = \frac{5}{0.05} = \mathbf{100 \text{ mA}}$$

Answer: (c) 100 mA. Option (d) would be correct if $\alpha_1 + \alpha_2 = 1$. Here $\alpha_1 + \alpha_2 = 0.95 < 1$, so the SCR does NOT latch; $I_A = 100 \text{ mA}$ with gate current maintained.

Q9. Without any gate signal ($I_G = 0$), the anode current is $I_A = (I_{CBO1} + I_{CBO2}) / (1 - \alpha_1 - \alpha_2)$. As V_{AK} increases, junction J2 begins to avalanche: the multiplication factor M increases both α_1 and α_2 . When $\alpha_1 + \alpha_2 = 1$ (at $V_{AK} = V_{BO}$), the denominator vanishes, $I_A \rightarrow \infty$, and the SCR fires. This is **forward voltage breakover** – undesirable and a rating violation. **Answer: (b)**

Q10. Total switching transition time per cycle: $t_{on} + t_{off} = 2 + 3 = 5 \mu\text{s}$. If the switching period must be at least 10 times this:

$$T_{s,min} = 10 \times 5 \mu\text{s} = 50 \mu\text{s} \Rightarrow f_{s,max} = \frac{1}{50 \times 10^{-6}} = \mathbf{20 \text{ kHz}}$$

Answer: (b) 20 kHz.

⚠ Pre-Print Corrections Required:

- Q6:** Change answer key from (c) to **(d) 120 W**. Revise option (c) to a different distractor (e.g. 80 W for t_{off} only at 20 kHz).
- Q7:** Revise options to (a) 60 W (b) 120 W (c) 180 W **(d) 240 W**. Answer: (d).
- Q8:** Answer key entry was mistyped as (d); correct answer is **(c) 100 mA**.

All conceptual questions Q1–Q5 and Q9–Q10 are correct as stated.

Dr. B. C. Roy Engineering College, Durgapur

Department of Electrical Engineering

CO-PO Mapping of Flip Learning Activity

Subject: Power Electronics (EE-504) Section: EE-2 Date: 1st September 2026

2 Course Outcome Addressed

CO1: Define and explain the characteristics and operation of power electronic devices and switching components.

Bloom's Level: BL2 (Understand) for Q1–Q5 | BL3 (Apply) for Q6–Q8 | BL4 (Analyse) for Q9–Q10

Topics: Ideal vs. Practical Switching Characteristics | Turn-On and Turn-Off Times | Switching Losses | Two-Transistor Model of SCR | Regenerative Turn-On Mechanism

References: M. H. Rashid, *Power Electronics*, PHI, 3rd Ed., Ch. 1 and Ch. 4; P. S. Bimbhra, *Power Electronics*, Khanna Publishers, 3rd Ed., Ch. 2–3.

NPTEL: Dr. G. Narayanan, IISc Bangalore – <https://nptel.ac.in/courses/108/108/108108035/>

3 CO-PO Mapping Table

CO	Course Outcome	Programme Outcomes (PO)											PSO		
		1	2	3	4	5	6	7	8	9	10	11	1	2	3
CO1	Define and explain characteristics and operation of power electronic devices	3	2	1	2	2	1	1	–	1	1	2	3	2	1

Table 1: CO1-PO/PSO Correlation (3: High, 2: Medium, 1: Low, –: No correlation), as per EE-504 lesson plan CO-PO mapping

4 Question-wise CO-PO Justification

Q	Skill / Concept Tested	CO	PO	BL
Q1	Identify the defining properties of an ideal switch and distinguish from practical device	CO1	PO1, PO2	BL2
Q2	Recall and explain the components of turn-on time (t_d , t_r) and their physical basis	CO1	PO1	BL2
Q3	Explain the physical origin of storage time t_s in minority-carrier devices and its impact on turn-off	CO1	PO1, PO2	BL2
Q4	Describe the cross-coupled connection in the two-transistor model and identify the feedback path	CO1	PO1, PO2	BL2
Q5	Explain the latching condition and justify why the gate pulse can be removed after SCR turn-on	CO1	PO1, PO2	BL2
Q6	Apply the switching loss formula to compute average power loss given device parameters and switching frequency	CO1	PO1, PO2	BL3
Q7	Apply the linear relationship between switching loss and frequency; predict loss at doubled frequency	CO1	PO1, PO2	BL3
Q8	Apply the two-transistor model I_A expression; compute anode current for given α values and gate current	CO1	PO1, PO2	BL3
Q9	Analyse the forward breakover mechanism; identify the condition for spontaneous firing and distinguish from gate-triggered firing	CO1	PO1, PO2	BL4
Q10	Analyse the practical frequency limit imposed by device switching times; apply the constraint to compute $f_{s,max}$	CO1	PO1, PO2	BL4

Table 2: Question-wise skill, CO, PO and Bloom's Level mapping

5 PO Description Reference

PO	Description (AICTE/NBA)
PO1	Engineering Knowledge: Apply knowledge of mathematics, science, and engineering fundamentals to explain and quantify device behaviour in power electronic circuits
PO2	Problem Analysis: Identify, formulate, and analyse device-level problems using switching-time, loss, and circuit model relationships

Table 3: Relevant PO descriptions for CO1 mapping

Dr. B. C. Roy Engineering College, Durgapur

Department of Electrical Engineering

Faculty Evaluation Report

Teaching-Learning Method: Flip Learning (FL)

Subject: Power Electronics (EE-504) Section: EE-2

Date: Tuesday, 1st September 2026 Time: 2:10–3:00 PM

Faculty: Kingsuk Majumdar, Ph.D. (Engg.) Dept. of EE, BCREC

6 Activity Details

Activity	Flip Learning (FL)
Subject / Code	Power Electronics / EE-504
Section	EE-2 (EE2X + EE2Y), B. Tech (EE), 5th Semester
Batch	2024–28
Session	2026–27 (ODD Semester)
Date and Time	Tuesday, 1st September 2026, 2:10–3:00 PM
Venue	_____
Pre-Class Note Shared On	30-August-2026 (Sunday, evening)
Total Students Present	_____ out of 47
Topics	Ideal vs. Practical Switching Characteristics Turn-On and Turn-Off Times Switching Losses Two-Transistor Model of SCR Regenerative Turn-On Mechanism
Module	Module 1 (Introduction) and Module 2 (PNPN Devices – Thyristors)
CO Addressed	CO1 – Define and explain characteristics and operation of power electronic devices
Bloom's Level	BL2 (Understand) for Q1–Q5 BL3 (Apply) for Q6–Q8 BL4 (Analyse) for Q9–Q10
Format	10 MCQ (GATE/ESE standard); 2 marks each; Total 20 marks; Negative marking $-\frac{2}{3}$

Table 4: Flip Learning Activity Details – EE-504

7 ? MCQ Summary

Section A – Conceptual (Q1–Q5):

- Q1: Ideal switch vs. practical switch properties
 Q2: Turn-on time components ($t_d + t_r$)
 Q3: Storage time t_s – minority carrier sweep-out
 Q4: Two-transistor model cross-coupling arrangement
 Q5: Gate pulse removal after SCR latch-on

Section B – Analytical / Application (Q6–Q10):

- Q6: Switching loss calculation at $f_s = 20$ kHz ($P_{sw} = 120$ W)
 Q7: Effect of doubling f_s on switching loss (240 W)
 Q8: Anode current from two-transistor model ($I_A = 100$ mA for $\alpha_1 + \alpha_2 = 0.95$)
 Q9: Forward breakover mechanism at V_{BO}
 Q10: Maximum switching frequency from device switching times ($f_{s,max} = 20$ kHz)

8 🕒 Phase-wise Time Observation

Phase	Duration	Time Slot	Faculty Observation
Individual MCQ	25 minutes	2:10–2:35 PM	_____
Consolidation	20 minutes	2:35–2:55 PM	_____
Doubt Clearance	5 minutes	2:55–3:00 PM	_____

Table 5: Phase-wise time and observation log

9 ⚠️ Common Mistakes / Misconceptions to Watch

Fill after conducting the activity. Typical mistakes to watch for:

- Q2 – t_{on} definition:** Students select option (a) (0% to 100% of I_{on}) instead of the standard 10%–90% definition. Emphasise that the 10%–90% convention matches the GATE/ESE data sheet definition. _____ (observed: Yes / No)
- Q3 – Storage time vs. fall time confusion:** Students confuse t_s (no current fall yet) with t_f (current actually falling). Minority carrier storage is a bipolar-device phenomenon; MOSFET has no t_s . _____ (observed: Yes / No)
- Q5 – Gate must be maintained:** Students select option (a) believing the gate pulse must be continuous. This is the most common conceptual error on SCR. The latching condition $\alpha_1 + \alpha_2 \geq 1$ makes gate unnecessary after turn-on. _____ (observed: Yes / No)

4. **Q6/Q7 – Switching loss linear in f_s :** Students do not recognise that $P_{sw} \propto f_s$. After computing Q6, Q7 should follow directly by doubling – if students recompute from scratch, it indicates they missed the proportionality. _____ (observed: Yes / No)
5. **Q8 – Latch-on confusion:** Students select (d) “theoretically unbounded” without checking whether $\alpha_1 + \alpha_2 \geq 1$. Here $\alpha_1 + \alpha_2 = 0.95$, which is less than 1, so the SCR does not latch and $I_A = 100$ mA. _____ (observed: Yes / No)
6. **Additional mistakes observed during class:**
- _____
- _____
- _____

10 Score Distribution

Score Range	16–20	10–15	5–9	<5
Number of Students				

Table 6: Score distribution (fill after evaluation)

Criterion	Excellent	Satisfactory	Needs Improvement
Pre-class preparation level (inferred from scores)			
Conceptual Q1–Q5 performance			
Analytical Q6–Q10 performance			
SCR two-transistor model questions (Q4, Q5, Q8, Q9)			
Switching loss questions (Q6, Q7)			
Active participation in consolidation phase			

Table 7: Class performance assessment summary (tick appropriate column)

11 Key Learning Points Emphasised During Consolidation

- The ideal switch is a theoretical baseline. Every practical device has conduction loss, leakage loss, and switching loss. Of these, switching loss is dominant at high frequencies and scales linearly with f_s – this is the single most important design constraint in modern power converters.
- Turn-on time ($t_d + t_r$) and turn-off time ($t_s + t_f$) are both non-zero in any real device. The storage time t_s is unique to minority-carrier devices (BJT, IGBT, SCR) and explains why they are slower to turn off than power MOSFETs.

3. The two-transistor model of the SCR is not merely a circuit trick – it directly explains three critical behaviours: (a) why a short gate pulse is sufficient to turn ON the device, (b) why the gate cannot turn OFF the device once latched, and (c) why excessive V_{AK} causes uncontrolled breakover firing.
4. The latch-on condition $\alpha_1 + \alpha_2 \geq 1$ is the threshold. Below it, the SCR is in its forward blocking state. At it, the device latches. Above it (due to avalanche at V_{BO}), the device fires without gate control. In normal operation, the gate pulse raises α_2 just enough to cross this threshold.
5. Additional points noted during class:

12 Faculty Remarks and Overall Assessment

Overall observation of pre-class preparation level:

Overall observation of in-class MCQ performance:

Effectiveness of Flip Learning methodology for this topic:

Action to be taken in next class (remedial / follow-up):

Kingsuk Majumdar, Ph.D. (Engg.)

Assistant Professor, Dept. of EE

Dr. B. C. Roy Engineering College, Durgapur

Date: _____

Signature: _____

13 NBA 2 TLM – Flip Learning Activity (EE-504)

13.1 Activity Overview

TLM Method:	Flip Learning (FL)
Subject / Code:	Power Electronics / EE-504
Section:	EE-2 (EE2X + EE2Y), B. Tech (EE), 5th Semester, Batch 2024–28
Date and Time:	Tuesday, 1st September 2026, 2:10–3:00 PM
Pre-Class Note Issued:	30-August-2026 (Sunday, evening) – 2 days before the activity
Faculty:	Kingsuk Majumdar, Ph.D. (Engg.), Dept. of EE, BCREC
Topics:	Ideal vs. Practical Switching Characteristics Turn-On and Turn-Off Times Switching Losses Two-Transistor Model of SCR Regenerative Turn-On Mechanism
Module:	Module 1 (Introduction) and Module 2 (PNPN Devices)
CO Addressed:	CO1 – Define and explain the characteristics and operation of power electronic devices and switching components
Bloom's Level:	BL2 (Understand) for Q1–Q5, BL3 (Apply) for Q6–Q8, BL4 (Analyse) for Q9–Q10
Total Students:	47 (EE2X: 24 students, EE2Y: 23 students)

13.2 Rationale for Topic Selection and Flip Format

The topics of ideal vs. practical switching characteristics and the two-transistor model of the SCR were selected for the Flip Learning activity for the following reasons. Both topics consist of structured descriptive and conceptual content – definitions, physical explanations, derivations from a circuit model, and a comparison table – that is well suited to independent pre-class reading from the prescribed textbook or study note, rather than requiring classroom derivation scaffolding. The Flip Learning format was chosen specifically because these topics span two modules (Module 1 and Module 2) that were in progress concurrently at the time of the activity (1st September 2026 falls in Week 5, during which Modules 1, 4, and 2 have all been partially or fully delivered). Moving the conceptual content to pre-class reading allowed the full 50-minute period to be used for GATE/ESE standard MCQ-based assessment (Phase 1), faculty-led consolidation of the most frequently misunderstood concepts (Phase 2), and doubt clearance (Phase 3). This serves the dual purpose of satisfying the NBA 2 TLM requirement and simultaneously reinforcing the theory component of Class Test-1 (scheduled 09-September-2026, covering Modules 1, 4, and 2).

13.3 Pre-Class Resource Provided

Sl.	Document	Format	Date Shared
1	Pre-class study note covering all 5 topics: ideal/practical switch, turn-on/turn-off times, switching losses, two-transistor model of SCR, regenerative turn-on mechanism. (EE504_FL_PreClass_Note_V2)	PDF	30-August-2026
2	NPTEL reference: Dr. G. Narayanan, IISc Bangalore, https://nptel.ac.in/courses/108/108/108108035/	URL	Shared with note

Table 8: Pre-class resources shared with students

13.4 Phase-wise Execution Plan

Phase	Duration	Time	Description
Individual MCQ	25 min	2:10–2:35 PM	Students attempt all 10 questions individually on the response sheet. No peer interaction permitted. Faculty circulates to ensure compliance and notes engagement level as a proxy for pre-class preparation.
Consolidation	20 min	2:35–2:55 PM	Faculty discusses each answer with correct reasoning, common error identification, and GATE/ESE examination tips. Special emphasis on: Q5 (gate not needed after latch), Q8 (check $\alpha_1 + \alpha_2$ before concluding latch-on), and Q9 (breakover vs. gate-triggered firing).
Doubt Clearance	5 min	2:55–3:00 PM	Open Q&A. Students may ask on any of the five topics. Faculty connects switching-time concepts to the upcoming Module 2 content on SCR triggering and commutation.

Table 9: Phase-wise execution plan for Flip Learning activity – EE-504

13.5 CO-PO Mapping Summary

CO	PO1	PO2	PSO1	PSO2	PSO3	Bloom's Level
CO1	3	2	3	2	1	BL2, BL3, BL4

Table 10: CO-PO-PSO correlation for Flip Learning activity (3: High, 2: Medium, 1: Low)

13.6 Supporting Documents for NBA File Submission

Sl.	Document	Format
1	Pre-class study note (EE504_FL_PreClass_Note_V2) with date of sharing (30-Aug-2026) as evidence	PDF
2	In-class MCQ question paper (Document 1 of this file, pages 1–3)	PDF
3	Student response sheets – minimum 10 samples	Single PDF
4	Score distribution sheet (filled Table in Faculty Evaluation Report, Section 3)	PDF
5	Photographs: students writing MCQ (GEO tag + non-GEO tag)	Single PDF
6	Faculty Evaluation Report (Document 3 of this file)	PDF
7	CO-PO mapping of activity (Document 2 of this file)	PDF

Table 11: Documents to be submitted for FL NBA 2 file – EE-504

Document Prepared by: Kingsuk Majumdar, Ph.D. (Engg.)

Department of Electrical Engineering
Dr. B. C. Roy Engineering College, Durgapur

Date: 30-August-2026 **Version:** V2

Copyright © 2026 Kingsuk Majumdar

 [GitHub](#)  [YouTube](#)